

# Jason Tsao

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## EDUCATION

### San Jose State University

San Jose, CA

*Bachelor of Science in Computer Engineering (GPA 3.70)*

*Aug. 2024 – Dec. 2027*

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Concepts & Methodology, Linear Algebra, Google Site Reliability Engineering

## TECHNICAL SKILLS

**Languages:** Python, Go, C, C++, SQL, JavaScript, TypeScript, Bash

**Frameworks & Data:** React, Node.js, Express, FastAPI, OpenCV, Prisma, SQLAlchemy, PostgreSQL, SQLite

**Platform & Orchestration:** Linux, Kubernetes, Helm, Docker, Terraform, Ansible, Gitlab, Calico, Cilium, MetalLB

**Infrastructure & Monitoring:** Proxmox, VMware ESX, Prometheus, Grafana, Nginx, Traefik

## EXPERIENCE

### IT Systems Administrator

May 2026 – Present

*Student Union, Inc. of SJSU*

*San Jose, CA*

- Implemented Kubernetes for containerized workloads, enabling continual scalable deployments for 40k students.
- Provisioned Gitlab/Terraform, configuring RBAC, remote state, and CI/CD workflows to streamline deployment.
- Configured Prometheus/Grafana across 50+ services, cutting incident resolution time by 30% via real-time alerts.
- Managed Proxmox VE, VMware ESX, backups, & high availability, ensuring 99.99% hardware service continuity.
- Built AI cloud agent to mitigate CVEs with MCP calls, reducing MTTR by 40% & saving 30 triage hours weekly.

### IT Technician

Feb. 2026 – May 2026

*Student Union, Inc. of SJSU*

*San Jose, CA*

- Boosted network throughput by 20% through resolving legacy VLAN, proxy, and firewall routing bottlenecks.
- Maintained MDF/IDF network capacity, managing 5+ switches and 240+ ports to ensure site connectivity.
- Engineered zero-touch autounattend.xml workflows for 120+ workstations, reducing deployment time by 60%.
- Standardized configuration management for 120+ devices via Intune and Jamf, achieving 100% compliance.

### Software Engineer, Development Team

Jan. 2026 – Present

*Software and Computer Engineering Society*

*San Jose, CA*

- Engineered a robust full-stack music streaming webui in TypeScript with React, Express, Prisma, and SQLite.
- Architected an egress-only WebSocket bridge for secure outbound connections, eliminating inbound firewalls.
- Containerized services via Docker Compose and deployed a systemd daemon piping yt-dlp media into mpv.

## PROJECTS

### StudyGuard | *Python, FastAPI, OpenCV, OpenAI Whisper, SQLAlchemy*

- Won at the Sseed Embodied AI Hackathon for building an innovative AI-powered focus-monitoring system.
- Built an OpenCV and GPT-4V pipeline that accurately classifies student focus states at 10-second intervals.
- Streamed a live camera feed directly to a FastAPI dashboard for real-time proctoring and focus visualization.
- Used a Whisper voice agent to remotely control Reachy Mini robot gestures and update the dnsmasq allowlist.

### Home Lab | *Docker, Ansible, Cloud-init, Nginx, Proxmox, WireGuard, pfSense, Ollama, Hermes*

- Configured pfSense as the primary router with segmented VLANs, custom firewall rules, and NAT policies.
- Leveraged cloud-init for initial bootstrapping and Ansible for idempotent, agentless configuration management.
- Integrated a Hermes agent via an Ollama API endpoint to automate complex internal infrastructure workflows.
- Implemented an Nginx reverse proxy with TLS termination to ensure secure and encrypted traffic routing.
- Established a WireGuard VPN tunnel to strictly enforce secure, zero-trust remote access to internal endpoints.